Friday 16 Oct: presenter's card

slides: `meeting_plots/oct16/slides.pdf`

Open with: "T1 is complete in 17 regions. I built the T3 test harness and finished T2 tier 1. The headline: terrain from elevation data predicts hiking speed in regions the model has never seen; OSM trail tags barely do. It survived every check I could think of."

Numbers to know

- **Data:** 17 regions, 2,876 hiking traces; walking set 1,992 outings, 290,640 segments of 50 m, 4,702 h.
- **Tobler:** 13% worse than our curve. Real hikers peak near **4.2 km/h**, not 6.
- **Unseen region** (skill vs slope curve): terrain **+2.1%** (trees +2.8%), tags +0.9% (+0.4%). Terrain beats tags in **16 of 17** regions.
- **Harder tests:** a whole zone left out +1.9%; West ↔ East +2.6%; newer hikes +2.2%; same at every GPS rate.
- **Whole hike:** time error 13.6% → 9.6% with the hiker's own pace (first km) + terrain.
- **Same hiker, same trail:** down is 8% faster than up on gentle grades, 22% on steep ones. Tobler says 42%.
- **Modes:** runners = walkers ×2.5; bikes need their own curve.

New since last week

- Why tags fail: not different meanings (within a hike, harder sac_scale is slower in every region). They're missing on much of the trail (1–96% by region).
- Sparse GPS traces read up to 11% slow: how we measure, not slower hikers.
- A 2026 paper (Huang et al., Hong Kong) uses boosted trees with weather, one city, random split: no transfer test.
- The 5-number formula travels less well than the full model (+0.6% for an unseen zone). Say so.

Ask Danni

1. Framing: "terrain transfers, tags don't". Agree?
2. Embeddings: keep as "helps locally, transfers less"?
3. Lidar understory: ∼12 GB for the White Mountains. Do it?
4. Data release: license (ODbL) and lab policy.
5. Cluster account: `engr-class-any` or the lab's?
6. Fix sparse traces in a T1 v2, or just report it?
7. Which 3–4 figures to keep (the draft is at 5 pages).

Same order as slide 14.

Hard questions, short answers

"Isn't +2% small?"
Per 50 m, yes: GPS noise and the hiker dominate. But it's consistent (17 of 17 regions in direction), its interval excludes zero, and it's more than twice what tags give.

"Is terrain just a stand-in for the region?"
No: it adds ∼1.9 points over each region's own average, and holds with a whole zone left out.

"Why no deep learning or images?"
About 2,000 outings is small. We tested frozen satellite embeddings: +1.1% in new regions, less reliable than terrain.

"What about weather?"
Date and time of day add nothing in new regions (seasons differ by region). Weather itself isn't tested yet.

"What did the AI do?"
Claude helped write and run the code and draft the documents; I review the work and I'm responsible for it.